

15 Ans: D

$$I \propto A^2 \text{ and } I \propto 1/r^2$$

Therefore, $A \propto 1/r$

$$\text{Hence } A_Q / A_P = r / (2r)$$

$$A_Q = 1/2 (8.0) = 4.0 \mu\text{m}$$

16 Ans: A